

SkinGenAI: Technical Project Documentation

Few-Shot Medical Image Lesion Segmentation Using Transfer Learning with MobileNetV3-UNet

1. Executive Summary & Project Specifications

Project Title	Few-Shot Medical Image Lesion Segmentation with Transfer Learning
System Identifier	SkinGenAI (Interactive Clinical Research Dashboard)
Model Architecture	MobileNetV3-Small (Encoder) + 4-Stage U-Net (Decoder with Skip Connections)
Core Technologies	PyTorch 2.11, Torchvision 0.26, Streamlit 1.55, Plotly, NumPy, Pillow
Dataset Benchmark	ISIC 2018 Task 1: Lesion Boundary Segmentation (Dermatoscopic RGB + Binary Masks)
Evaluated Splits	50, 100, 250, and 500 images (Low-Data / Few-Shot Regimes)
Best Validation Metrics	Dice: 87.07% IoU: 77.52% (at 500 images, Epoch 9)
Remote Repository	https://github.com/Gowthami1214/SkinGenAI

2. Clinical Background & Research Motivation

Malignant melanoma represents the most fatal form of skin cancer, yet early surgical excision yields a 5-year survival rate exceeding 99%. Clinical assessment relies on dermoscopy and the **ABCD criteria** (Asymmetry, Border irregularity, Color variegation, Diameter). Automated lesion boundary segmentation is the foundational first step: without an exact border, computerized systems cannot compute asymmetry, border irregularity, or pigment network distribution.

The Data Scarcity Dilemma: Standard deep learning segmentation models require thousands of pixel-level annotations. Manual pixel contouring by certified dermatologists takes 3–8 minutes per image, creating an insurmountable bottleneck for rare conditions or resource-limited health systems. SkinGenAI investigates: *What is the minimum labeled training set needed to produce clinically viable segmentation masks using transfer learning?*

3. Deep Learning Architecture: MobileNetV3-UNet

The model integrates a lightweight **MobileNetV3-Small** backbone (ImageNet-pretrained) with a symmetrical **U-Net decoder** via multi-scale skip connections. This design pairs high parameter efficiency (~1.5M parameters) with spatial boundary reconstruction.

Pipeline Stage	Feature Dim	Channels	Architectural Role
Input Dermoscopy	224 × 224	3 (RGB)	Standardized normalized dermoscopic photo
Encoder Stage 0 (Skip 4)	112 × 112	16	Fine hair, edge, and contrast transitions
Encoder Stage 1 (Skip 3)	56 × 56	16	Local pigment networks and texture maps
Encoder Stage 2 (Skip 2)	28 × 28	24	Regional lesion symmetry features
Encoder Stage 8 (Skip 1)	14 × 14	48	Lesion global shape context
Bottleneck Bridge	7 × 7	576	Highest semantic abstraction
Decoder UpConv 1–4	7→14→28→56→112	48, 24, 16, 16	ConvTranspose2d upsampling + Skip fusion
Final Projection	224 × 224	1 logit	Sigmoid → Binary thresholding at $\theta=0.50$

4. Optimization & Validation Metrics

Hybrid Loss: Dermoscopy images exhibit severe foreground/background class imbalance (lesions cover only 15–30% of canvas area). SkinGenAI trains with $L_{total} = L_{BCE} + L_{Dice}$, where BCE stabilizes per-pixel probability convergence and Dice Loss directly penalizes missed lesion areas.

Metric	Formula	Clinical Interpretation
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Dice Similarity (DSC)	$2 \cdot TP / (2 \cdot TP + FP + FN)$	Gold standard ISIC metric. Measures contour completeness and overlap harmonic mean.
IoU (Jaccard Index)	$TP / (TP + FP + FN)$	Strict overlap area divided by total union area. Mathematically equal to Dice / (2 - Dice).
Precision	$TP / (TP + FP)$	Measures over-segmentation. High precision ensures minimal healthy skin is excised.
Recall (Sensitivity)	$TP / (TP + FN)$	Critical oncology metric. High recall guarantees malignant tumor cells are not missed.

5. Empirical Benchmark Results across Few-Shot Configurations

Configuration n	Training Samples	Best Epoch	Validation Dice	Validation IoU	Empirical Progression
Config 1	50 images	15	0.5104 (51.0%)	0.3621 (36.2%)	Baseline: Identifies dark centroids; rough borders
Config 2	100 images	15	0.5765 (57.6%)	0.4106 (41.1%)	+6.6% Gain: Learns radial symmetry & artifact suppression
Config 3	250 images	15	0.8246 (82.5%)	0.7081 (70.8%)	Critical Inflection Point (+24.8% jump)
Config 4	500 images	9	0.8707 (87.1%)	0.7752 (77.5%)	Near-supervised SOTA: Rapid convergence by epoch 9

Key Research Finding: An exponential performance jump occurs between **100 and 250 images** (Dice increases from 57.65% to 82.46%). At 250 images, the dataset provides sufficient pigment diversity for MobileNetV3's multi-scale skip connections to fully engage, proving that **250 samples represent the critical viability threshold** for clinical transfer learning in dermoscopy.

6. Interactive Streamlit Dashboard System

- **Executive Dashboard:** Status badge, 4-step pipeline visualizer, live validation score cards, and color-coded Dice progress bars.
- **Experiment Lab:** Hyperparameter inspection, Plotly learning curves (Data Size vs Dice/IoU), and data insights.
- **Segmentation Studio:** Sample ISIC library or upload, dynamic threshold slider (θ : 0.10–0.90), 4-panel visualizer (Original, Heatmap, Mask, Overlay), mask export button, and side-by-side comparative panel across all 4 models.
- **Model Comparison:** Score cards, interactive **Dice Coefficient Matrix Heatmap**, side-by-side grouped bar charts, and radar fingerprints.
- **Test Evaluation:** Discovers image-mask pairs in sample_data/ to compute live Dice, IoU, Precision, and Recall on the fly.
- **About Project:** Technical methodology, architectural diagrams, and clinical disclaimers.

7. Cloud Deployment & Academic Disclaimer

Continuous Cloud Deployment: The codebase is deployed on **Streamlit Community Cloud** connected to GitHub repository Gowthami1214/SkinGenAI on branch main. Dependencies in requirements.txt utilize PyTorch's CPU extra-index-url, reducing cloud image build sizes from 2.5GB to ~170MB.

Clinical Disclaimer: SkinGenAI is an academic research prototype designed to study few-shot transfer learning under data scarcity. Predictions are not certified by the FDA, CE, or medical regulatory bodies, and must never be used for primary diagnostic purposes without a licensed dermatologist.